

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester of B. Tech Examination (IT)
May 2016
IT214: Data Structures & Files

Date: 24.05.2016, Tuesday

Time: 10:00 am to 1:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Do as directed. [07]

(1) In sorting of following input using **selection sort**, what will be the output string **01**

after **3rd pass**? Input Sequence = 25, 17, 31, 13, 2.

- (A) 2, 13, 17, 25, 31 (B) 2, 13, 17, 31, 25
(C) 2, 13, 25, 17, 31 (D) None of the above

(2) Is below statement true or false. **01**

“Depth-first search is used to find connected component of a graph.”

(3) Give the correct matching for the following pairs. **02**

- | | |
|--------------------|-------------------------------------|
| (a) Balance Factor | (1) Balance Factor = 0, -1, 1 |
| (b) AVL | (2) Recursion |
| (c) Preorder | (3) H _L - H _R |
| (A) a-3, b-2, c-1 | (B) a-3, b-1, c-2 |
| (C) a-1, b-2, c-3 | (D) a-1, b-3, c-2 |

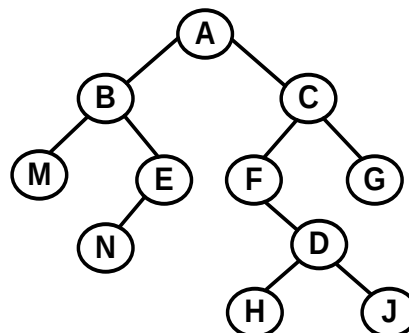
(4) Explain the impact of data structure on performance of algorithm with the help of suitable example. **03**

Q-2 [14]

(a) Define graph. Explain adjacency matrix and adjacency list storage representation of a graph. **04**

(b) Answer following questions. (Any Two) 10

1 Draw Full Threaded Binary Tree for given binary tree.



- 2 Apply Linear and Binary Search on Following Data : 1,2,3,4,5,6,7,8,9,10 Search Keys are : a) 1 b) 5 c) 11. Compare no. of Comparisons required for Linear and Binary Search. Write the general formula to find maximum no. of comparison required for binary search and linear search.

- 3 From given orders draw the correct tree. Find Post-order traversal.

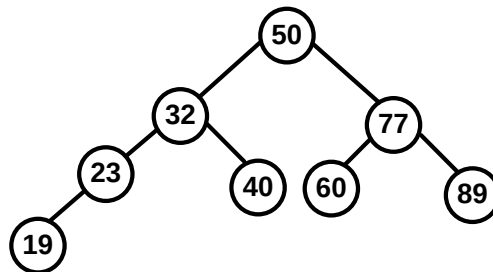
Preorder: **K C A B F G J O X P U Z**

Inorder: **A B C F G J K O P U X Z**

Q-3 Attempt Any Two.

[14]

- (a) Construct the Binary Search Tree using following data. Show each steps.
32, 45, 12, 11, 13, 92, 78, 66, 17, 70, 98, 108.
Delete 92 and then 32 from the tree. Draw tree after each deletion.
- (b) What is the basic idea of Insertion sort? Apply Insertion sort on below string and count no. of comparison and no. of exchanges in each pass.
Input Sequence : 40, 2, 1, 43, 3, 65, 0, -1, 58, 3, 42, 4
- (c) Insert 45, 12, 25 and 39 into the following AVL Tree. Draw balanced tree after each insertion.



SECTION – II

Q-4 Do as directed.

[07]

- (1) State whether following statement is TRUE or FALSE. **02**
- (a) Doubly linked list increase the search time with compare to array.
 - (b) Queue concepts can be implemented using linked list.
 - (c) Collision means different hash key but same data.
 - (d) Recursion uses queue structure for implementation.
- (2) Insertion and deletion operations can be made at either end of the structure is known as _____. **01**
- (3) DFS uses _____ data structure and BFS uses _____ data structure. **01**
- (4) What is the purpose to convert infix expression into reverse polish notation? **01**
- (5) For given array of float A [-100:52, 60:89] find the total number of elements. **02**
Assume that Base address is 5001. Find address of A [0, 71] element, if it is stored in row major order.

Q-5

[14]

- (a) How many minimum moves are required to move 18 disks from source to destination? If 2 milliseconds are required to move one disk then calculate total time required to move 14 disks from source to destination towers. **04**

(b) Answer following questions.(Any Two)

10

(1) Evaluate the following expressions using stack:

i) $4\ 3\ /\ 2\ 1\ 3\ +\ 2\ -\ * \ 6\ /\ +$ ii) $+ \ /\ /\ * +\ 3\ 1\ +\ 4\ 2\ 6\ 2\ 3$

(2) Arrange the following data in the array using hashing. Use linear probing to resolve collisions. Array size is 8 (index starts from 0)

Data: **50, 47, 19, 21, 11, 45, 6, 25**

Hash function $H(x) = ((x^2) \bmod 5) + 3$

(3) Convert the following infix expression into its equivalent reverse polish notation using stack.

$$(((a + b) / (c + d) * e) + f) + (g + e) ^ h$$

Q-6 Answer following questions.(Any Two)

[14]

(a) Create a Doubly linked list using following data that arranges data in descending order. Show list after each insertion.

Addresses: 9020, 6110, 8990, 5011, 6229. Elements : 15, 11, 9, 13, 18

(b) Show circular queue contents with front and rear after each step with size=5.

(a) Insert 10, 20, 30 (b) Delete (c) Insert 40, 50, 60, 70. Initially queue is empty.

(c) Create a max heap tree and sort the given values in ascending order using heap sort
4, -12, 16, 32, 24, 57, -11, 69, 96.
